

Project #3 (Dichotomous Contrasts)

Step #5 w/ target scale:

```
Console Terminal Background Jobs
R 4.5.1 ~ /Brownlee/
> proj3A$W_NegAtt_GB <- rowSums(proj3A[,listWA])
> proj3A$H_NegAtt_GB <- rowSums(proj3A[,listHA])
> proj3A$D_NegAtt_GB <- proj3A$W_NegAtt_GB - proj3A$H_NegAtt_GB
> model3 <- lm(data=proj3A, proj3A$D_NegAtt_GB ~ 1)
> summary(model3)

Call:
lm(formula = proj3A$D_NegAtt_GB ~ 1, data = proj3A)

Residuals:
    Min       1Q   Median       3Q      Max
-23.8405  -4.8405  -0.8405   5.1595  22.1595

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)    2.841      0.487   5.832 1.64e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.808 on 256 degrees of freedom

> |
```

Step #6 w/ target scale:

```
Console Terminal Background Jobs
R 4.5.1 ~ /Brownlee/
> model4 <- lm(data=proj3A, proj3A$W_NegAtt_GB ~ 1)
> summary(model4)

Call:
lm(formula = proj3A$W_NegAtt_GB ~ 1, data = proj3A)

Residuals:
    Min       1Q   Median       3Q      Max
-12.3852  -4.3852  -0.3852   3.6148  15.6148

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  12.3852     0.4049  30.59  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.492 on 256 degrees of freedom

> model5 <- lm(data=proj3A, proj3A$H_NegAtt_GB ~ 1)
> summary(model5)

Call:
lm(formula = proj3A$H_NegAtt_GB ~ 1, data = proj3A)

Residuals:
    Min       1Q   Median       3Q      Max
-9.5447  -4.5447  -0.5447   3.4553  18.4553

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   9.5447     0.3587  26.61  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.751 on 256 degrees of freedom

> |
```

Step #7 w/ target scale:

There was a significant difference between wives and husbands ($d = .46$, $t(256) = 5.83$, $p < .001$), with wives receiving higher negative attributes scores ($M = 12.39$, $SD = 6.49$) than husbands ($M = 9.54$, $SD = 5.75$). This represents a small effect.

Step #8 w/ target scale:

1. Screen shot(s). This section should show necessary screen shots with the correct commands and results for an analysis (including t -values, means, and standard deviations).	Possible points 3	Deductions 0	Score 3														
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G. Summary lists the necessary means and standard deviations. The value for any mean (or standard deviation) is always placed <u>after</u> the reader has been given the name of the variable.	2
H. If an effect is significant, the summary accurately interprets the effect size as trivial, small, medium, or large.	2
I. The summary is easy to read. Everything is in the order listed above or in some other sensible order. It is formatted correctly. It does not contain unnecessary information. It does not contain sentences that are difficult to understand.	2
J. Regardless of whether the information is correct, the summary at least includes information listed for d , and t , and if results are significant, it includes a statement about the magnitude of effect.	2
K. The p -value is correctly formatted. If a p -value is less than .001, it should be printed as " $p < .001$ " in the text and never printed as " $p = .000$."	1

3. Attachments.

Possible points	Deductions	Score
0.25	0	.25

Criterion	Deduction
A. This grading table has been completed and is attached.	0.25

Step #9 (Step #5 repeated w/ 2nd scale):

```

Console Terminal x Background Jobs x
R 4.5.1 ~ /Brownlee/
> proj3A$W_OwnSoftEmo_GB <- rowSums(proj3A[,listWOSE])
> proj3A$H_OwnSoftEmo_GB <- rowSums(proj3A[,listHOSE])
> proj3A$D_OwnSoftEmo_GB <- proj3A$W_OwnSoftEmo_GB - proj3A$H_OwnSoftEmo_GB
> model7 <- lm(data=proj3A, proj3A$D_OwnSoftEmo_GB ~ 1)
> summary(model7)

Call:
lm(formula = proj3A$D_OwnSoftEmo_GB ~ 1, data = proj3A)

Residuals:
    Min       1Q   Median       3Q      Max
-11.0311  -3.0311  -0.0311   2.9689  10.9689

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    2.0311     0.2581   7.869 1.01e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.138 on 256 degrees of freedom

> |

```

Garrett Brownlee
PSY 4400 01
Dr. Sanford
10/18/25

Step #9 (Step #6 repeated w/ 2nd scale):

```
Console Terminal Background Jobs
R 4.5.1 · ~/Brownlee/

> model8 <- lm(data=proj3A, proj3A$W_OwnSoftEmo_GB ~ 1)
> summary(model8)

Call:
lm(formula = proj3A$W_OwnSoftEmo_GB ~ 1, data = proj3A)

Residuals:
    Min       1Q   Median       3Q      Max
-9.7782 -2.7782  0.2218  3.2218  6.2218

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   9.7782     0.2432   40.2    <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.899 on 256 degrees of freedom

> model9 <- lm(data=proj3A, proj3A$H_OwnSoftEmo_GB ~ 1)
> summary(model9)

Call:
lm(formula = proj3A$H_OwnSoftEmo_GB ~ 1, data = proj3A)

Residuals:
    Min       1Q   Median       3Q      Max
-7.7471 -3.7471  0.2529  3.2529  8.2529

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   7.7471     0.2599   29.81    <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.167 on 256 degrees of freedom

> |
```

Step #9 (Step #7 repeated w/ 2nd scale):

There was a significant difference between wives and husbands ($d = .50$, $t(256) = 7.87$, $p < .001$), with wives receiving lower own soft emotion scores ($M = 9.78$, $SD = 3.90$) than husbands ($M = 7.75$, $SD = 4.17$). This represents a medium effect.

Step #9 (Step #8 repeated w/ 2nd scale):

4. **Screen shot(s).** This section should show necessary screen shots with the correct commands and results for an analysis (including *t*-values, means, and standard deviations).

Possible points	Deductions	Score
3	0	3

Criterion	Deduction
F. Section does not contain any content errors (missing steps or incorrect results).	2
G. Section contains no more than two content errors.	1
H. Screenshot size and resolution allow it to be read easily.	2
I. Screenshot cropped and does not show command lines with errors.	1
J. Correctly followed instructions to create scales (in Part A only).	3

5. **Written summary.** This section includes a written summary of a dichotomous contrast analysis.

Possible points	Deductions	Score
12	0	12

Criterion	Deduction
L. The summary demonstrates an understanding of the basic results and does not include any major mistakes or misinterpretations (such as a mix up about whether results are significant, a contradictory statement implying that results are both significant and not significant, or a completely wrong interpretation).	6
M. The first sentence needs to use the words “difference between” and accurately describe what the difference is between. This statement should only use the word “groups” if the analysis is an independent samples analysis (and not a paired samples analysis). It should NOT describe the difference as a difference between two things that are not being compared in this study (for example, the analysis never tests a difference between two variables).	2
N. Summary reports the correct <i>d</i>	2
O. Summary reports the correct <i>t</i> -value and degrees of freedom	2
P. Summary reports the correct <i>p</i> value	2
Q. Summary clarifies the nature of the observed difference by naming an outcome variable and stating what had higher values and what had lower values.	2
R. Summary lists the necessary means and standard deviations. The value for any mean (or standard deviation) is always placed <u>after</u> the reader has been given the name of the variable.	2
S. If an effect is significant, the summary accurately interprets the effect size as trivial, small, medium, or large.	2

	T. The summary is easy to read. Everything is in the order listed above or in some other sensible order. It is formatted correctly. It does not contain unnecessary information. It does not contain sentences that are difficult to understand.	2
	U. Regardless of whether the information is correct, the summary at least includes information listed for d , and t , and if results are significant, it includes a statement about the magnitude of effect.	2
	V. The p -value is correctly formatted. If a p -value is less than .001, it should be printed as " $p < .001$ " in the text and never printed as " $p = .000$."	1

6. Attachments.		
Possible points	Deductions	Score
0.25	0	.25

Criterion	Deduction
B. This grading table has been completed and is attached.	0.25

Step #13 w/ target scale:

```

Console Terminal x Background Jobs x
R 4.5.1 · ~/Brownlee/
> Model10 <- lm(data=proj3B, CSI ~ break_GB)
> summary(Model10)

Call:
lm(formula = CSI ~ break_GB, data = proj3B)

Residuals:
    Min       1Q   Median       3Q      Max
-3.2708 -0.5208  0.2292  0.7292  1.6600

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  5.27083    0.07221  72.993  < 2e-16 ***
break_GB    -0.93083    0.19430  -4.791 3.47e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9019 on 179 degrees of freedom
Multiple R-squared:  0.1136,    Adjusted R-squared:  0.1087
F-statistic: 22.95 on 1 and 179 DF,  p-value: 3.474e-06
>

```

Step #14 w/ target scale:

```
Console Terminal x Background Jobs x
R 4.5.1 ~ /Brownlee/

$`0`

call:
lm(formula = CSI ~ 1, data = w)

Residuals:
    Min       1Q   Median       3Q      Max
-3.2708 -0.5208  0.2292  0.7292  0.9792

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   5.27083    0.06601   79.84  <2e-16 ***
---
Signif. codes:
  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8245 on 155 degrees of freedom

$`1`

call:
lm(formula = CSI ~ 1, data = w)

Residuals:
    Min       1Q   Median       3Q      Max
 -2.09  -1.34   0.16   1.16   1.66

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    4.340      0.259   16.76 9.48e-15 ***
---
Signif. codes:
  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.295 on 24 degrees of freedom

>
```

Step #15 w/ target scale:

There was a significant difference between the couples in stable relationships and the couples in dissolving relationships ($d = 1.03$, $t(179) = -4.79$, $p < .001$), with the couples in stable relationships reporting higher relationship satisfaction scores ($M = 5.27$, $SD = 0.82$) than the couples in dissolving relationships ($M = 4.34$, $SD = 1.30$). This represents a large effect.

Step #16 w/ target scale:

7. **Screen shot(s).** This section should show necessary screen shots with the correct commands and results for an analysis (including *t*-values, means, and standard deviations).

Possible points	Deductions	Score
3	0	3

Criterion	Deduction
K. Section does not contain any content errors (missing steps or incorrect results).	2
L. Section contains no more than two content errors.	1
M. Screenshot size and resolution allow it to be read easily.	2
N. Screenshot cropped and does not show command lines with errors.	1
O. Correctly followed instructions to create scales (in Part A only).	3

8. **Written summary.** This section includes a written summary of a dichotomous contrast analysis.

Possible points	Deductions	Score
12	0	12

Criterion	Deduction
W. The summary demonstrates an understanding of the basic results and does not include any major mistakes or misinterpretations (such as a mix up about whether results are significant, a contradictory statement implying that results are both significant and not significant, or a completely wrong interpretation).	6
X. The first sentence needs to use the words “difference between” and accurately describe what the difference is between. This statement should only use the word “groups” if the analysis is an independent samples analysis (and not a paired samples analysis). It should NOT describe the difference as a difference between two things that are not being compared in this study (for example, the analysis never tests a difference between two variables).	2
Y. Summary reports the correct <i>d</i>	2
Z. Summary reports the correct <i>t</i> -value and degrees of freedom	2
AA. Summary reports the correct <i>p</i> value	2
BB. Summary clarifies the nature of the observed difference by naming an outcome variable and stating what had higher values and what had lower values.	2
CC. Summary lists the necessary means and standard deviations. The value for any mean (or standard deviation) is always placed <u>after</u> the reader has been given the name of the variable.	2
DD. If an effect is significant, the summary accurately interprets the effect size as trivial, small, medium, or large.	2
EE. The summary is easy to read. Everything is in the order listed above or in some other sensible order. It is formatted correctly. It does not	2

contain unnecessary information. It does not contain sentences that are difficult to understand.	
FF. Regardless of whether the information is correct, the summary at least includes information listed for d , and t , and if results are significant, it includes a statement about the magnitude of effect.	2
GG. The p -value is correctly formatted. If a p -value is less than .001, it should be printed as " $p < .001$ " in the text and never printed as " $p = .000$."	1

9. **Attachments.**

Possible points	Deductions	Score
0.25	0	.25

Criterion	Deduction
C. This grading table has been completed and is attached.	0.25

Step #17 (Step #13 repeated w/ 2nd scale):

```

Console  Terminal x Background Jobs x
R 4.5.1 · ~/Brownlee/
> Model12 <- lm(data=proj3B, anxious ~ break_GB)
> summary(Model12)

call:
lm(formula = anxious ~ break_GB, data = proj3B)

Residuals:
    Min       1Q   Median       3Q      Max
-2.7116 -0.9544 -0.3244  1.0456  3.0456

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.2044     0.1094  29.303  <2e-16 ***
break_GB       0.7572     0.2942   2.574   0.0109 *
---
Signif. codes:
  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.366 on 179 degrees of freedom
Multiple R-squared:  0.03568,    Adjusted R-squared:  0.03029
F-statistic: 6.623 on 1 and 179 DF,  p-value: 0.01088
> |
  
```

Step #17 (Step #14 repeated w/ 2nd scale):

```
Console Terminal x Background Jobs x
R 4.5.1 · ~/Brownlee/
$ `0`

call:
lm(formula = anxious ~ 1, data = w)

Residuals:
    Min       1Q   Median       3Q      Max
-2.2044 -0.9544 -0.3244  1.0456  3.0456

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.2044      0.1102   29.07  <2e-16 ***
---
Signif. codes:
  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.377 on 155 degrees of freedom

$ `1`

call:
lm(formula = anxious ~ 1, data = w)

Residuals:
    Min       1Q   Median       3Q      Max
-2.7116 -0.8316  0.0384  0.7884  2.7884

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.9616      0.2584   15.33 6.75e-14 ***
---
Signif. codes:
  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.292 on 24 degrees of freedom

>
```

Step #17 (Step #15 repeated w/ 2nd scale):

There was a significant difference between the couples in stable relationships and the couples in dissolving relationships ($d = -0.55$, $t(179) = 2.57$, $p < .05$), with the couples in stable relationships reporting lower anxious romantic attachment scores ($M = 3.20$, $SD = 1.38$) than the couples in dissolving relationships ($M = 3.96$, $SD = 1.29$). This represents a medium effect.

Step #17 (Step #16 repeated w/ 2nd scale):

10. Screen shot(s). This section should show necessary screen shots with the correct commands and results for an analysis (including t -values, means, and standard deviations).	Possible points	Deductions	Score
	3	0	3

Criterion	Deduction
P. Section does not contain any content errors (missing steps or incorrect results).	2
Q. Section contains no more than two content errors.	1
R. Screenshot size and resolution allow it to be read easily.	2
S. Screenshot cropped and does not show command lines with errors.	1
T. Correctly followed instructions to create scales (in Part A only).	3

11. Written summary. This section includes a written summary of a dichotomous contrast analysis.	Possible points	Deductions	Score
	12	0	12

Criterion	Deduction
HH. The summary demonstrates an understanding of the basic results and does not include any major mistakes or misinterpretations (such as a mix up about whether results are significant, a contradictory statement implying that results are both significant and not significant, or a completely wrong interpretation).	6
II. The first sentence needs to use the words “difference between” and accurately describe what the difference is between. This statement should only use the word “groups” if the analysis is an independent samples analysis (and not a paired samples analysis). It should NOT describe the difference as a difference between two things that are not being compared in this study (for example, the analysis never tests a difference between two variables).	2
JJ. Summary reports the correct d	2
KK. Summary reports the correct t -value and degrees of freedom	2
LL. Summary reports the correct p value	2

MM. Summary clarifies the nature of the observed difference by naming an outcome variable and stating what had higher values and what had lower values.	2
NN. Summary lists the necessary means and standard deviations. The value for any mean (or standard deviation) is always placed <u>after</u> the reader has been given the name of the variable.	2
OO. If an effect is significant, the summary accurately interprets the effect size as trivial, small, medium, or large.	2
PP. The summary is easy to read. Everything is in the order listed above or in some other sensible order. It is formatted correctly. It does not contain unnecessary information. It does not contain sentences that are difficult to understand.	2
QQ. Regardless of whether the information is correct, the summary at least includes information listed for d , and t , and if results are significant, it includes a statement about the magnitude of effect.	2
RR. The p -value is correctly formatted. If a p -value is less than .001, it should be printed as " $p < .001$ " in the text and never printed as " $p = .000$."	1

12. Attachments.		
Possible points	Deductions	Score
0.25	0	.25

Criterion	Deduction
D. This grading table has been completed and is attached.	0.25